CAPTCHA Inspector - Feature Documents (one file per module)

Technology Brainz assignment: import a CSV of Contact Us pages, identify the CAPTCHA on each, and make sure CAPTCHAs do not stop the workflow.

What "handle" means in this tool (read first)

The tool DOES	The tool does NOT
Open the page in headless Chromium (Playwright)	Solve or bypass reCAPTCHA / hCaptcha / Turnstile
Detect and classify the CAPTCHA (provider, type, widget state)	Click, tick or interact with a live CAPTCHA
Save a screenshot and evidence	Submit any form or harvest tokens
Route every CAPTCHA to a **human review queue** so the batch keeps going	Work around robots.txt or access blocks
Solve **our own synthetic text CAPTCHAs** with OCR (OCR Laboratory)	

Reason: solving or bypassing third-party CAPTCHAs is against their terms and would be unauthorised access. The workflow is not blocked: a CAPTCHA is identified, logged and queued, and the batch moves on to the next site.

Modules

#	File	Screen
1	01_URL_INSPECTOR.md	URL Inspector
2	02_SCAN_HISTORY.md	Scan History
3	03_CAPTCHA_RESULTS.md	CAPTCHA Results
4	04_HUMAN_REVIEW.md	Human Review
5	05_OCR_LABORATORY.md	OCR Laboratory
6	06_VERIFICATION.md	Verification
7	07_CSV_BATCH_MANAGER.md	CSV Batch Manager
8	08_REPORTS.md	Reports
9	09_EVALUATION_AND_OVERVIEW.md	Evaluation and Overview
10	10_SETTINGS_AND_SECURITY.md	Settings, login, security

Typical flow: URL Inspector -> Scan History -> CAPTCHA Results -> Human Review. For many sites: CSV Batch Manager -> Reports.

1. URL Inspector

Purpose: inspect one website. You give a URL; the tool opens it, finds contact/support pages and forms, and detects CAPTCHAs.

How to use

1. Open `/inspector`.
2. Enter the Target Website URL (a contact-page URL inspects that page; a home page lets the tool discover contact pages).
3. Set Max Pages (1-20) and Crawl Depth (0-3). Use 1 and 0 for a single page.
4. Tick *Allow Localhost* only for your own local test site.
5. Click **Start Live Inspection**.

What happens inside

- Checks the URL against the allow-list and SSRF rules (private/internal addresses are refused).
- Checks robots.txt; stops if the site disallows.
- Loads each page in Chromium, waits for it to settle, reads the rendered DOM, scripts and iframes.
- Detects Google reCAPTCHA, hCaptcha, Cloudflare Turnstile and others; records subtype and widget state.
- Counts forms, saves a screenshot, stores everything in the database.

What you see

- Cards: Crawled Pages, Contact Touchpoints, CAPTCHA Detected, Crawl Duration.
- "Discovered Pages & Evidence": one row per page with provider, forms, handling (`human_review` or `no_challenge`) and an **Inspect** button (screenshot + details).

Failure behaviour

If no page could be read (for example a timeout) the scan is **Failed** with the reason. It is never reported as "no CAPTCHA". Page timeout is 30 s (`CRAWLER_PAGE_TIMEOUT_MS`).

2. Scan History

Purpose: permanent record of every inspection ever run. This is where you see what was detected and of which type.

How to use

1. Open `/scans`.
2. Search by URL, domain or scan id. Filter: All / Completed / Crawling / Failed. Use refresh for new scans.
3. Click **Inspect** on a row to open that scan's pages and screenshots.

Columns

Target, scan id, status, pages crawled, contact pages found, CAPTCHA providers found, duration, time.

Reading a row

- **Completed + provider name** (for example Google reCAPTCHA): CAPTCHA identified.
- **Completed + None Detected + pages > 0**: pages were read and no CAPTCHA was found.
- **Failed**: the page could not be read (timeout, blocked, unreachable). The reason is shown. This is not evidence that the site has no CAPTCHA.

3. CAPTCHA Results

Purpose: a focused list of the pages where a CAPTCHA was found, with its type.

How to use

1. Open `/results`.
2. Keep "CAPTCHA Detections Only" on to hide clean pages.
3. Click **Inspect** to see the screenshot and evidence.

What it shows

Page path and title, HTTP status, forms found, provider badge (Google reCAPTCHA / hCaptcha / Turnstile), handling strategy (`human_review`), review status.

Confidence shows "-" because per-page confidence is not measured; the tool does not invent a number.

4. Human Review (how CAPTCHAs are "handled")

Purpose: the queue of CAPTCHAs that a person must decide on. This is what keeps the workflow moving: the scanner never waits on a CAPTCHA; it records it here and continues.

How to use

1. Open `/reviews`.
2. Click **Review Now** on an item. See the page URL, provider, challenge type, reason and screenshot.
3. Mark Approved / Rejected / Manually verified and add notes.

Guarantees

- Each decision is stored with who made it and when (audit trail).
- The system never solves, clicks or bypasses the CAPTCHA itself.
- Confidence shows "-" when unknown (never "NaN%").

5. OCR Laboratory (the only place CAPTCHAs are actually solved)

Purpose: demonstrate automated solving on **our own synthetic text CAPTCHAs**. It does not apply to reCAPTCHA, hCaptcha or Turnstile.

How to use

1. Open `/ocr`.
2. **Run OCR Evaluation:** generates a distorted text image, runs the computer-vision pipeline and a text reader, shows the answer, correct/incorrect and timing.
3. **Randomize:** new random text.
4. **Engine / Split / Samples + Run Benchmark:** measures accuracy over many generated images.

Note

The reported accuracy (about 99.3%) is a benchmark on our own generator. It is not a claim about real-world CAPTCHAs.

6. Verification

Purpose: send a token to the provider's official verification service using the providers' published **test** keys, and record the real answer.

How to use

1. Open `/verification`.
2. Choose a provider and submit the test token.
3. Read the result in "Recorded attempts".

Outcomes recorded

verified, failed, already used, missing token, provider unavailable. A success is never faked. No production keys are used.

7. CSV Batch Manager (the employer's main workflow)

Purpose: import the 200-row CSV (`USA_Ecommerce_Platforms_200_Contact_Pages.csv`) and scan every Contact Us page.

How to use

1. Open `/batch` .
2. **Choose CSV**, then **Import & Validate CSV**. The summary shows total / valid / invalid rows, columns preserved and row-numbered errors.
3. Tick "Automatically start batch scan", or click **Launch Batch Scan**.
4. Watch the results table fill in. Click **Details** for robots status, widget state, expected value and screenshot.
5. Download the reports.

For each site the tool

1. Checks robots.txt (stops on disallow, 401/403/429, challenge, 5xx).
2. Waits between requests to the same host (pacing).
3. Opens the Contact Page, detects the CAPTCHA, saves a screenshot.
4. Queues CAPTCHAs for human review and continues with the next site.
5. Compares the detected CAPTCHA with the CSV's "Captcha" column.

Expected vs Detected

Match, Mismatch, Not detected, Not accessible (blocked or unreachable), Not verified (no expected value). The percentage uses only inspected rows that have an expected value, and the sample size is shown.

Honest expectation

Many real sites block automated access. Those rows show "Not accessible"; they are not forced.

8. Reports

Purpose: download results.

How to use

1. Open `/reports`.
2. See past batches and counters (total batches, CAPTCHAs documented, verified clean sites).
3. **Export CSV** for the audit report; the batch screen also offers Expected-vs-Detected as CSV or Markdown.

Safety

Cells starting with `=`, `+`, `-`, `@`, tab or CR are prefixed with `'` so spreadsheets cannot run injected formulas.

9. Evaluation and Overview

Purpose: totals and charts computed from what was actually run: scans, pages, CAPTCHA types, OCR runs, verification attempts, errors and review cases.

How to use

Open `/` (Overview) and `/evaluation`. With no data the screens show zero or empty. Nothing is invented or mocked.

10. Settings, login and security

Login (`/login`)

Role-based accounts (admin / operator). Passwords are never stored in plain text.

Settings (`/settings`)

Read-only view of backend status, crawler limits, OCR engine and policies. Admins also see the tamper-evident audit trail. Local file paths are not exposed.

Security controls

- Allow-list of domains; SSRF protection (private and internal addresses refused, redirects re-validated, DNS pinned).
- robots.txt respected; no access-control bypass.
- Live scanning is off unless `ENABLE_LIVE_SCAN=true` .
- Never clicks CAPTCHAs, never submits forms, never harvests tokens.